Check for updates

aps ASSOCIATION FOR PSYCHOLOGICAL SCIENCE

Current Directions in Psychological Science
2021, Vol. 30(5) 418–424
© The Author(s) 2021
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/09637214211031939
www.psychologicalscience.org/CDPS
SAGE

Perception, Action, and Intrinsic Motivation in Infants' Motor-Skill Development

Daniela Corbetta
Department of Psychology, The University of Tennessee, Knoxville

Abstract

Perception, action, and intrinsic motivation play an essential role in early development, promoting the creation and refinement of new and more complex forms of behaviors as infants try a range of sensorimotor patterns in their environment. I use the example of infants' reaching to illustrate how goal-directed action emerges from the intersection of seemingly distinct visual and proprioceptive-tactile-motor spaces that form in the early months following birth. The intersection of these two spaces begins with a casual contingent event involving vision and action: when the hand happens to contact a target. This event, which marks the onset of reaching, provides new behavioral value, reinforces the motor action, and intrinsically motivates infants to attempt to reproduce the behavior. Subsequent repeated cycles of perception and action lead to the exploration of a range of motor responses and a progressive alignment of the visual space with the proprioceptive-tactile-motor space, ultimately fostering the selection and refinement of increasingly successful and refined reaching patterns. Extensive hands-on experience in the environment and learning about the immediate outcomes of actions play a critical role in shaping behavioral development.

Keywords

infant development, perception, action, intrinsic motivation, reinforcement learning

Perception and action are fundamental to everyday life. As individuals perceive and act, they ensure continuous exchanges with the environment and promote conditions for learning that are essential for acquiring new skills or fine-tuning existing ones. In adults, perception and action work closely together in most daily activities. Whether one manipulates an object or speaks to another person, these actions are carried out in concert with crucial ongoing sensory information originating from the body (via proprioception generated by muscles and tendons) and picked up from the environment (via vision, touch, and/or audition). The continuous pairing of multisensory information with the motor system helps orchestrate behavior and contributes to regulate and attune responses to the environment daily, moment by moment.

At birth, newborns are endowed with the basic senses of touch, hearing, sight, smell, and taste, which are crucial to keep them connected to their surroundings. They also produce spontaneous movements, which generate proprioceptive feedback essential to develop body awareness and perceive body position and movements. However, unlike adults, infants need to discover what they can do with their body, need to figure out the space surrounding them, and need to learn how to relate their movements to perceived outer information to develop behaviors adapted to their physical and social world. In this essay, I describe how newborns' initial rudimentary sensorimotor responses develop into a visual space and a proprioceptive-motor space that appear to follow separate tracks over the first few months of life. During this period, infants acquire greater perceptual capabilities and accumulate extensive basic sensorimotor experiences essential for the formation of future goal-directed behaviors. Then, I take the example of reaching for objects, one of the earliest goal-directed behaviors to form, to illustrate

Corresponding Author:
Daniela Corbetta, Department of Psychology, The University of Tennessee, Knoxville
Email: dcorbett@utk.edu

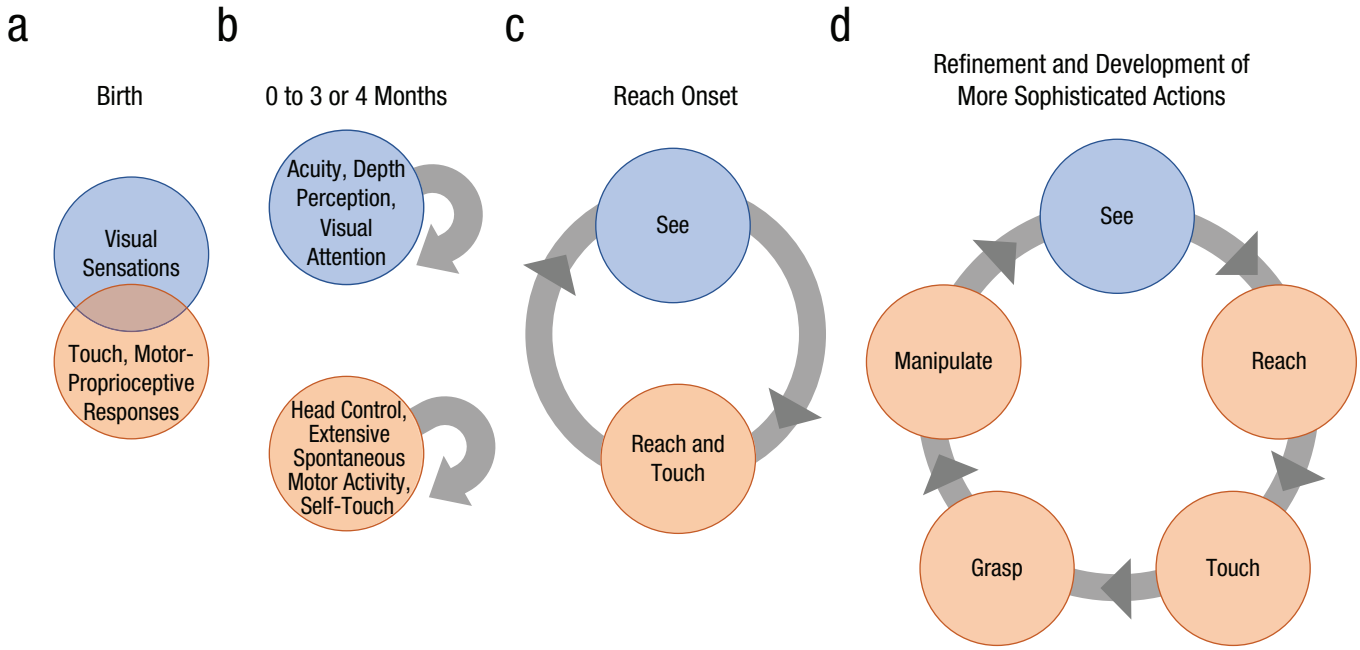


Fig. 1. Developmental steps leading to infants' reaching and its refinement. At birth (a), rudimentary visual and touch sensations are closely interconnected with rudimentary motor responses. From 0 to 3 or 4 months of age (b), infants develop seemingly separate sensory and motor spaces through daily experience: a visual space that is far-reaching and a proprioceptive-tactile-motor space limited to the body and arm movements. At the onset of reaching (c), the visual and proprioceptive-motor spaces intersect, and initial intrinsically motivated perception-action cycles lead to the alignment of vision to the motor space. Finally (d), the repeated cycles of perception and action are progressively refined and differentiated, which leads to the development of grasping and more sophisticated actions.

how a goal-directed behavior initially emerges from an accidental event that brings together the visual and proprioceptive-tactile-motor spaces and creates a novel outcome. Sensory feedback from this outcome gives infants new incentives, direction, and intrinsic motivation to reproduce the movement that generated it. From that moment, repeated cycles of perception and action play an instrumental role in aligning the visual and sensorimotor spaces and in promoting the refinement of the behavior. Perceptual and motor exploration through trial and error, mediated by intrinsic motivation and selective reinforcement, is an important mechanism through which progressive behavioral attunements to the environment occur.

Sensorimotor Responses at Birth

From birth, newborns display rudimentary responses reflecting tight interconnections between the sensory and motor systems (Fig. 1a). These responses include newborns' primitive reflexes. For example, touching specific skin areas, such as the mouth, cheek, or palm, will trigger specific motor reactions, such as sucking, turning the head, or grasping. Newborns who are just a few days old also display motor reactions in response to visual inputs. For example, they generate leg patterns similar to the stepping reflex when held in the air while exposed to dots of lights flowing toward them (Barbu-Roth et al., 2009). They can also produce a rudimentary form of reaching called *prereaching*, which entails a full arm and hand extension, sometimes directed toward a perceived object, when stable head support is provided (Grenier, 1981; von Hofsten, 1982). And they can move their arm to bring their hand into view when supine if the conditions allow it (van der Meer, 1997). These initial patterns of sensorimotor interconnections are important to boost development; however, additional behavioral components and sensorimotor experiences are necessary to promote the development of more complex and adaptive behaviors. In the next section, I use the example of infants' reaching to illustrate how aspects of vision and sensorimotor experiences first develop along seemingly separate tracks, to provide important foundations for the emergence of goal-directed behavior.

Toward the Emergence of Goal-Directed Reaching

For a long time, learning to reach was considered a gradual process of coordination between the eyes and the hands, culminating around 3 to 5 months of age

with the eyes guiding the hand toward the object to contact it (Piaget, 1936/1952; White et al., 1964). In recent years, however, it has become clear that the development of reaching is the product of a more complex process entailing the formation and integration of many more behavioral components (see Corbetta, Wiener, & Thurman, 2018, for a review). Several observations suggest that the visual and motor experiences that develop during this early period follow somewhat separate tracks before these tracks interconnect again at the onset of reaching (Fig. 1b).

Nearly 30 years ago, von Hofsten (1984) was the first to document that the initial prereaching responses of neonates change into arm extensions that are no longer accompanied by a hand opening or visual fixations on the target. He reported that arm extensions with hand openings while fixating the target progressively return at around 10 weeks of age, in preparation for reaching.

Other behavioral components not necessarily functional at birth also need to develop to promote the emergence of reaching. Postural control, for example, and particularly head control, which emerges around 2 months of age, is critical for the development of reaching. During the same period, newborns' visual acuity sharpens, and depth perception develops. These, in turn, contribute to increase infants' attentional capabilities. Very quickly, infants can see beyond the limits of their body and scrutinize the broader and more distant space surrounding them.

In the months preceding the onset of reaching, infants also accumulate extensive self-generated sensorimotor experiences. In a recent study, my colleagues and I (DiMercurio et al., 2018) observed infants' arm movements weekly during their first 2 months of life while awake and supine. We found that infants produced an astonishingly high rate of spontaneous movements and spontaneous touches to their body and the surrounding supporting surface (> 200 hand contacts over a 10-min time window). These touches were distributed all over the areas the infants could attain within arm's reach, and this frequency of touches remained high over the 2-month period of observation. These arm movements and touches were produced in complete absence of visual guidance. We rarely observed instances when infants stopped moving to look at their hands. Thus, over the many hours of wake time that infants typically spend in a supine position during their early months, it appears that visual and manual activities might rarely overlap. The considerable proprioceptive, motor, and tactile experience infants gain may provide crucial information about their body and arm movements in space, but none of this experience appears to be monitored by vision.

From these observations, it seems that in the months before the onset of reaching, infants discover and experience two distinct sensory spaces that are scaled very differently: a visual space that grows rapidly with increased acuity, that progressively expands beyond arm's length, and that is far-reaching and a more restricted proprioceptive-tactile sensorimotor space defined by the body and the limited space within which arm movements can be deployed. Some of our work (Corbetta et al., 2014) suggests that despite changes in all of these precursory behaviors, the intersection of the proprioceptive-motor and visual sensory spaces happens only after infants have made their first hand-target contact, and not before as proposed in the past (e.g. White et al., 1964; see Corbetta, Wiener, Thurman, & McMahon, 2018, for a review).

Indeed, in a study in which my colleagues and I used eye tracking for the first time to readily examine how eye-hand coordination formed during the transition to goal-directed reaching (Corbetta et al., 2014), we found, contrary to prior observations, that before and after the onset of reaching, infants' vision was poorly aligned with the motor response and not fully predictive of where infants reached. It took infants 5 weeks of repeated reaching experience for their vision to align with the motor response and become somewhat more predictive, a finding opposite to the classic view that vision guides the hand to the target. Our observations were consistent with other findings (Clifton et al., 1993; Thelen et al., 1993) in that infants initially relied more on their sense of proprioception than on their vision to bring their arm in contact with the objects, which suggests that learning to reach may require a calibration, or scaling down, of the visual sensory space to align it with the proprioceptive-tactile-motor space. In the next section, I examine how the visual and proprioceptive-motor spaces come together and how intrinsic motivation and repeated cycles of perception and action lead to the consolidation and refinement of infants' reaching.

Learning to Reach Through Perception, Action, Intrinsic Motivation, and Contingent Reinforcement

When 3- to 4-month-old infants approach the onset of reaching, they stare intensively at an object located within reach while actively flailing their arms, but fail to contact it (Corbetta et al., 2014; Corbetta, Wiener, & Thurman, 2018; Thelen et al., 1993). Despite preparatory signs such as increased hand opening, infants seem to have no idea how to intercept the object. My colleagues and I proposed that infants on the brink of

developing a new skill operate under conditions of uncertainty, as they have never had the specific experience of practicing the skill in its functional form (Corbetta, DiMercurio, et al., 2018). Infants do not know how to use their arms to contact the object, and they do not know how to match proprioceptive information with perceived visual information.

The first experience that provides infants with some incentive to proceed with reaching is a casual event, as infants' hand path happens to cross the area where an object is located. That moment, when the hand contacts the target, provides contingent multisensory feedback in which vision, touch, and proprioception are all perceived at once in time and space and can be traced back to an arm movement and arm posture that led to them. Infants are not able to reproduce the exact same movement right away, but this event provides a reinforcing value to the action performed, sparking renewed interest and a motivation to try to contact the object again (Williams, Corbetta, & Cobb, 2015). The process that unfolds is one of trial and error, involving perception and action; each new behavioral attempt sets the stage for future new attempts (Fig. 1c) At this early stage, infants can only reach and touch an object with their hand. It takes 3 to 4 weeks for grasping to develop (von Hofsten & Lindhagen, 1979). Nonetheless, these early cycles of perception and action promote the exploration of a wide range of movement attempts directed toward the target, some successful and some unsuccessful. The outcomes of these attempts, paired with contingent sensory feedback, result in a progressive calibration of the visual space to the proprioceptive-tactile-motor space and selection of increasingly more direct reaching trajectories.

This process can be viewed from the perspective of reinforcement learning (Sutton & Barto, 2018). The term "reinforcement learning" was coined in the fields of computer sciences and machine learning to capture a process by which intelligent systems autonomously and dynamically discover and learn new behaviors in an unsupervised manner via initially random and self-produced interactions within complex and uncertain environments. Reinforcement learning has been used to model learning of many skills (e.g., Moulin-Frier et al., 2014), including infants' reaching (Berthier et al., 2005), and is neuroscience inspired (Gottlieb & Oudeyer, 2018). Central to the reinforcement-learning process are intrinsically motivated exploratory actions aimed at maximizing and exploiting the rewards they produce. Positive rewards are important because they provide reinforcing value to the actions taken and increase the probabilities that those rewarded actions are going to be selected and performed again in future attempts.

Reward is associated with the release of a neurotransmitter known as dopamine in an area of the mesolimbic brain area called the nucleus accumbens. Although there are many hypotheses regarding the rewarding properties of dopamine, it has consistently been identified as having motivational functions in motor learning and as reinforcing response-reward associations (Wise, 2004). The value assigned to actions via intrinsic rewards has also been associated with the pruning and selection of neural networks (Sporns & Edelman, 1993). Thus, when infants first accidentally contact a visually attended object with their hand, they receive rewarding reinforcement that motivates them to attempt to reproduce the behavior. Infants can take advantage of their prior sensorimotor experience to guide their arm extensions toward the approximate target location. As they do so, they can bring vision into alignment with the motor response. Once aligned, vision progressively takes the lead in planning future movements. From the first moment infants contact a target with their hand, the frequency of successful reaches increases by the day (Williams & Corbetta, 2016; Williams, Corbetta, & Guan, 2015). The hand trajectories, initially highly variable, become straighter over successive attempts (Thelen et al., 1996). More complex repeated cycles of seeing, reaching, touching, and grasping form, and they contribute to refining the behavior (Fig. 1d), as goal-directed movements begin to be more precisely scaled to objects' features (Corbetta & Snapp-Childs, 2009).

Beyond Reaching

This process of discovering new action patterns through exploration of an initial context of random movements and uncertainty has been observed in infants' development of a variety of skills. For instance, 3-month-old infants can learn specific patterns of leg kicking or even identify precise leg flexions to activate a mobile above their head (Angulo-Kinzler, 2001; Angulo-Kinzler et al., 2002; Thelen, 1994). Around the age of 4 months, they learn to differentiate arm movements in response to contingent reinforcement (Jacquey et al., 2020). And when infants are about 8 months old, they are able to learn how to coordinate their hands to retrieve an object concealed in a box (Bojczyk & Corbetta, 2004). At later ages, perception and action can organize patterns of tool manipulation, promote the discovery of hidden properties of objects, and even support the learning of object-label mappings (Corbetta, DiMercurio, et al., 2018). Thus, from initial action encounters with

the environment and through contingent feedback, infants become intrinsically motivated to try new movement patterns to reproduce rewarding outcomes. Cycles of perception and action, via trial and error, are the main vehicles through which varied movement solutions are discovered, explored, and perfected.

The environmental contexts and perceptual information available play an undeniable role in sensorimotor learning and can lead to variations in developmental trajectories. For example, in a study using the object-retrieval task, 6-month-old infants presented with a transparent box discovered how to retrieve a concealed object and learned how to coordinate their hands to do so efficiently much faster than infants presented with an opaque box (Bojczyk & Corbetta, 2004).

Also, work on sticky-mittens training has highlighted the importance of tactile feedback for learning to reach in infancy (Corbetta et al., 2016; Williams, Corbetta, & Guan, 2015). The sticky-mittens paradigm was created by Needham et al. (2002) to provide simulated grasping to young infants who had not yet developed the skill. However, the mittens' effectiveness for learning to reach has been questioned (van den Berg & Gredebäck, 2020; Williams, Corbetta, & Guan, 2015). My colleagues and I explored this issue in a study with infants who were 2 months and 21 days old and not capable of reaching. Infants were trained with nonprickly Velcro mittens that were soft to the touch and that covered the palm of their hands. Touching an object with the Velcro-covered mittens caused the object to stick to the mittens, as in Needham et al.'s studies. However, our mittens were altered such that infants' fingers were uncovered and could directly touch the object at contact. We also trained another group of infants with similar open-finger mittens that would not allow the object to stick at contact, and we had a control group that did not receive training. After 14 days of training, only the nonsticky-mittens group outperformed the control group in their rate of object contact; the sticky-mittens group did not (Williams, Corbetta, & Guan, 2015). Subsequent analyses of the data from the sticky-mittens group revealed that only 2 infants from that group increased their rate of reaching over the 14 days; they were the ones who made direct finger contact with the object. Infants who did not make such contact did not increase their rate of reaching (Corbetta et al., 2016). Thus, what drove learning to reach was not the provision of simulated grasping via sticky mittens, but rather the direct skin contact with the object that provided crucial tactile feedback. This finding underscores the intrinsically motivating and reinforcing role of sensory feedback in infants' motor learning.

In a subsequent study, we investigated how enhanced sensory consequences of movement fostered the development of reaching in nonreaching infants not wearing mittens (Williams & Corbetta, 2016). We used objects that moved and sounded when touched to enhance action-contingent feedback. We found that the increase in the rate of reaching responses was even greater with these enhanced objects than in the sticky-mittens group from our previous study (Williams, Corbetta, & Cobb, 2015).

Concluding Remarks

Perception, action, and intrinsic motivation play an essential role in the formation and the shaping of infants' actions. When infants are learning new skills, perception and action allow them to capitalize on their prior sensorimotor experiences and promote multisensory integration. The new forms of behavior that emerge and their reinforcing sensory consequences provide new incentives to reproduce the actions and further practice the behaviors, thus allowing for a refinement of the motor responses and their greater attunement to the environment. Extensive hands-on experience in the world and learning about the immediate outcomes of actions play a critical role in shaping development.

Recommended Reading

Corbetta, D., DiMercurio, A., Wiener, R. F., Connell, J. P., & Clark, M. (2018). (See References). A more detailed account of the perception-action loop in infant development and its impact on the learning of various motor and cognitive skills.

Corbetta, D., Thurman, S. L., Wiener, R. F., Guan, Y., & Williams, J. L. (2014). (See References). A report on a longitudinal study on the transition to reaching in infancy that shows how vision aligns with the motor response.

Corbetta, D., Wiener, R. F., & Thurman, S. L. (2018). (See References). A review of all the behavioral elements that matter for learning to reach in infancy.

Gottlieb, J., & Oudeyer, P.-Y. (2018). (See References). A review of intrinsically motivated sampling of the world in autonomous active learning from a neuroscience and computational perspective.

Wise, R. A. (2004). (See References). A review of the role of dopamine in motor learning and motivation.

Transparency

Action Editor: Robert L. Goldstone
Editor: Robert L. Goldstone
Declaration of Conflicting Interests
The author(s) declared that there were no conflicts of interest with respect to the authorship or the publication of this article.

ORCID iD

Daniela Corbetta https://orcid.org/0000-0002-1403-9171

References

Angulo-Kinzler, R. M. (2001). Exploration and selection of intralimb coordination patterns in 3-month-old infants. *Journal of Motor Behavior*, *33*(4), 363–376. https://doi.org/10.1080/00222890109601920

Angulo-Kinzler, R. M., Ulrich, B., & Thelen, E. (2002). Three-month-old infants can select specific leg motor solutions. *Motor Control*, *6*(1), 52–68.

Barbu-Roth, M., Anderson, D. I., Desprès, A., Provasi, J., Cabrol, D., & Campos, J. J. (2009). Neonatal stepping in relation to terrestrial optic flow. *Child Development*, *80*(1), 8–14. https://doi.org/10.1111/j.1467-8624.2008.01241.x

Berthier, N. E., Rosenstein, M. T., & Barto, A. G. (2005). Approximate optimal control as a model for motor learning. *Psychological Review*, *112*(2), 329–346. https://doi.org/10.1037/0033-295X.112.2.329

Bojczyk, K. E., & Corbetta, D. (2004). Object retrieval in the 1st year of life: Learning effects of task exposure and box transparency. *Developmental Psychology*, *40*(1), 54–66. https://doi.org/10.1037/0012-1649.40.1.54

Clifton, R. K., Muir, D. W., Ashmead, D. H., & Clarkson, M. G. (1993). Is visually guided reaching in early infancy a myth? *Child Development*, *64*(4), 1099–1110. https://doi.org/10.2307/1131328

Corbetta, D., DiMercurio, A., Wiener, R. F., Connell, J. P., & Clark, M. (2018). How perception and action fosters exploration and selection in infant skill acquisition. In J. M. Plumert (Ed.), *Advances in child development and behavior: Vol. 55. Studying the perception-action system as a model system for understanding development* (pp. 1–29). Academic Press.

Corbetta, D., & Snapp-Childs, W. (2009). Seeing and touching: The role of sensory-motor experience on the development of infant reaching. *Infant Behavior and Development*, *32*(1), 44–58. https://doi.org/10.1016/j.infbeh.2008.10.004

Corbetta, D., Thurman, S. L., Wiener, R. F., Guan, Y., & Williams, J. L. (2014). Mapping the feel of the arm with the sight of the object: On the embodied origins of infant reaching. *Frontiers in Psychology*, *5*, Article 576. https://doi.org/10.3389/fpsyg.2014.00576

Corbetta, D., Wiener, R. F., & Thurman, S. L. (2018). Learning to reach in infancy. In D. Corbetta & M. Santello (Eds.), *Reach-to-grasp behavior: Brain, behavior, and modelling across the life span* (pp. 18–41). Taylor & Francis.

Corbetta, D., Wiener, R. F., Thurman, S. L., & McMahon, E. (2018). The embodied origins of infant reaching: Implications for the emergence of eye-hand coordination. *Kinesiology Review*, *7*(1), 10–17. https://doi.org/10.1123/kr.2017-0052

Corbetta, D., Williams, J. L., & Haynes, J. M. (2016). Bare fingers, but no obvious influence of "prickly" Velcro! In the absence of parents' encouragement, it is not clear that "sticky mittens" provide an advantage to the process of learning to reach. *Infant Behavior and Development*, *42*, 168–178. https://doi.org/10.1016/j.infbeh.2015.05.001

DiMercurio, A., Connell, J. P., Clark, M., & Corbetta, D. (2018). A naturalistic observation of spontaneous touches to the body and environment in the first 2 months of life. *Frontiers in Psychology*, *9*, Article 2613. https://doi.org/10.3389/fpsyg.2018.02613

Gottlieb, J., & Oudeyer, P.-Y. (2018). Towards a neuroscience of active sampling and curiosity. *Nature Reviews Neuroscience*, *19*(12), 758–770. https://doi.org/10.1038/s41583-018-0078-0

Grenier, A. (1981). La "motricité libérée" par fixation manuelle de la nuque au cours des premières semaines de la vie [Released motor skills by manual fixation of the neck during the first weeks of life]. *Archives Françaises de Pédiatrie*, *38*, 557–562.

Jacquey, L., Popescu, S. T., Vergne, J., Fagard, J., Esseily, R., & O'Regan, K. (2020). Development of body knowledge as measured by arm differentiation in infants: From global to local? *British Journal of Developmental Psychology*, *38*(1), 108–124. https://doi.org/10.1111/bjdp.12309

Moulin-Frier, C., Nguyen, S. M., & Oudeyer, P.-Y. (2014). Self-organization of early vocal development in infants and machines: The role of intrinsic motivation. *Frontiers in Psychology*, *4*, Article 1006. https://doi.org/10.3389/fpsyg.2013.01006

Needham, A., Barrett, T., & Peterman, K. (2002). A pick-me-up for infants' exploratory skills: Early simulated experiences reaching for objects using 'sticky mittens' enhances young infants' object exploration skills. *Infant Behavior and Development*, *25*(3), 279–295. https://doi.org/10.1016/S0163-6383(02)00097-8

Piaget, J. (1952). *The origins of intelligence in children* (M. Cook, Trans.). Basic Books. (Original work published 1936)

Sporns, O., & Edelman, G. M. (1993). Solving Bernstein's problem: A proposal for the development of coordinated movement by selection. *Child Development*, *64*(4), 960–981.

Sutton, R. S., & Barto, A. G. (2018). *Reinforcement learning: An introduction*. MIT Press.

Thelen, E. (1994). Three-month-old infants can learn task-specific patterns of interlimb coordination. *Psychological Science*, *5*(5), 280–285. https://doi.org/10.1111/j.1467-9280.1994.tb00626.x

Thelen, E., Corbetta, D., Kamm, K., Spencer, J. P., Schneider, K., & Zernicke, R. F. (1993). The transition to reaching: Mapping intention and intrinsic dynamics. *Child Development*, *64*(4), 1058–1098. https://doi.org/10.2307/1131327

Thelen, E., Corbetta, D., & Spencer, J. P. (1996). Development of reaching during the first year: Role of movement speed. *Journal of Experimental Psychology: Human Perception and Performance*, *22*(5), 1059–1076. https://doi.org/10.1037/0096-1523.22.5.1059

van den Berg, L., & Gredebäck, G. (2020). The sticky mittens paradigm: A critical appraisal of current results and explanations. *Developmental Science*. Advance online publication. https://doi.org/10.1111/desc.13036

van der Meer, A. L. (1997). Keeping the arm in the limelight: Advanced visual control of arm movements in neonates.

European Journal of Paediatric Neurology, *1*(4), 103–108. https://doi.org/10.1016/S1090-3798(97)80040-2

von Hofsten, C. (1982). Eye–hand coordination in the newborn. *Developmental Psychology*, *18*(3), 450–461. https://doi.org/10.1037/0012-1649.18.3.450

von Hofsten, C. (1984). Developmental changes in the organization of prereaching movements. *Developmental Psychology*, *20*(3), 378–388. https://doi.org/10.1037/0012-1649.20.3.378

von Hofsten, C., & Lindhagen, K. (1979). Observations on the development of reaching for moving objects. *Journal of Experimental Child Psychology*, *28*(1), 158–173. https://doi.org/10.1016/0022-0965(79)90109-7

White, B. L., Castle, P., & Held, R. (1964). Observations on the development of visually-directed reaching. *Child Development*, *35*(2), 349–364. https://doi.org/10.2307/1126701

Williams, J. L., & Corbetta, D. (2016). Assessing the impact of movement consequences on the development of early reaching in infancy. *Frontiers in Psychology*, *7*, Article 587. https://doi.org/10.3389/fpsyg.2016.00587

Williams, J. L., Corbetta, D., & Cobb, L. (2015). How perception, action, functional value, and context can shape the development of infant reaching. *Movement and Sport Sciences/Science et Motricité*, *89*, 5–15. https://doi.org/10.1051/sm/2015005

Williams, J. L., Corbetta, D., & Guan, Y. (2015). Learning to reach with "sticky" or "non-sticky" mittens: A tale of developmental trajectories. *Infant Behavior and Development*, *38*, 82–96. https://doi.org/10.1016/j.infbeh.2015.01.001

Wise, R. A. (2004). Dopamine, learning and motivation. *Nature Reviews Neuroscience*, *5*(6), 483–494. https://doi.org/10.1038/nrn1406